

B.Tech. Degree VI Semester Examination April 2014**ME 601 INSTRUMENTATION AND CONTROL SYSTEMS**
(2006 Scheme)

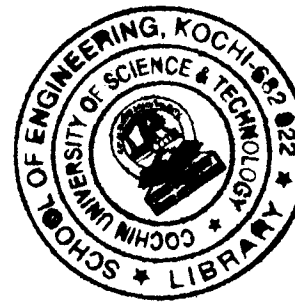
Time : 3 Hours

Maximum Marks : 100

PART A
(Answer *ALL* questions)

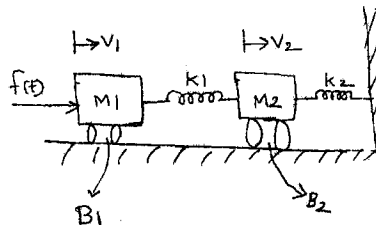
(8 × 5 = 40)

- I. (a) What do you mean by zero order instruments? Explain with example.
 (b) Draw the functional block diagram of a measurement system.
 (c) State the five laws of thermo couples.
 (d) Explain the working of an optical pyrometer with neat diagram.
 (e) Define transfer function.
 (f) Explain force voltage analogy.
 (g) Explain Nyquist criteria.
 (h) Write a short note on DC servomotor.

**PART B**

(4 × 15 = 60)

- II. (a) Explain the method of corrections of spurious inputs. (10)
 (b) Explain the step response of a 1st order instruments. (5)
- OR**
- III. Classify the different errors in a measurement system. Explain with examples. (15)
- IV. Explain the different types of strain gauges. Derive an expression for gauge factor. (15)
- OR**
- V. (a) Explain the working of ORSAT's apparatus with neat diagram. (10)
 (b) Explain the working principle of RTD. (5)
- VI. (a) Derive expressions for different time domain specifications. (12)
 (b) What are the different static error constants? (3)
- OR**
- VII. For the mechanical system shown draw the force voltage and force current analogous circuits. (15)



- VIII. The open loop transfer function of a unity feed back system is (15)

$$G(s) = \frac{1}{s(1+0.2s)(1+0.02s)}$$
 Draw bode plot and determine gain margin and phase margin.

OR

- IX. (a) State and explain Routh Hurwitz criterion? (5)
 (b) A certain unity feed back system has the following open loop transfer function. (10)

$$G(s)H(s) = \frac{k}{s(s+1)(s+3)}$$
 Draw root locus.